

Problem Set 1

POLS W4732

Fall 2026

Due September 18 before the TA section

Instructions. For each formal claim, include two or three algebraic inequalities or best-response comparisons that establish it. Unless directly following notation from slides or the text of the problem, define your notation.

Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1 (Sequential matching pennies). *Two students submit different formalizations of the sequential matching-pennies game from class. Student A includes only actions chosen at nodes that are reached in equilibrium. Student B includes a complete contingent plan for every decision node but assigns payoffs directly to actions rather than action profiles.*

1. Identify every error in each formalization.
2. Write the corrected player set, strategy sets, and payoff functions.
3. Change one feature of the verbal description so that the players no longer have equally many pure strategies, and formalize the resulting game.

Exercise 2 (Iterated deletion). *Consider the game below, except that Player 1's payoff at (D, R) is the parameter $x \in \mathbb{R}$.*

		Player 2	
		<i>L</i>	<i>R</i>
Player 1	<i>U</i>	2, 1	-1, 2
	<i>M</i>	0, 3	0, 1
	<i>D</i>	-1, -2	x , -1

Characterize, for every x , the actions that survive iterated deletion of strictly dominated actions. Prove your characterization for each region of x by identifying, at every deletion step, the deleted action, an action that strictly dominates it, and the payoff inequalities that establish dominance. Evaluate the claim that the answer is unchanged if “strictly” is replaced by “weakly,” and exhibit any difference in the surviving game.

Exercise 3 (Generalized Rock–Paper–Scissors). A win with Rock, Paper, or Scissors is worth r , p , or s , respectively; a loss is worth -1 , and a tie is worth zero. Assume $r, p, s > 0$.

		Player 2		
		R	P	S
Player 1	R	0, 0	$-1, p$	$r, -1$
	P	$p, -1$	0, 0	$-1, s$
	S	$-1, r$	$s, -1$	0, 0

Assume $p = s = 1$ and $r > 0$. Derive the completely mixed equilibrium and verify that every probability is positive and sums to one. Determine what happens as $r \rightarrow 0$. A proposed solution claims that a more valuable winning action must be played more often. Evaluate the claim and explain the strategic logic.

Exercise 4 (The Calvert–Wittman model). Two candidates L and R compete in an election. The winner gets to choose a policy $p \in [0, 1]$. Candidates L and R only care about the implemented policy—not about winning per se—and have ideal policies 0 and 1. Before the election, they announce platforms p_L and p_R in $[0, 1]$. There is a single voter with ideal point $m \in (0, 1)$. The closer platform to m wins the election; a tie leads to the implementation of $p_{\text{tie}} = \tau p_L + (1 - \tau) p_R$.

1. Write both payoff functions $u_L(p_L, p_R)$ and $u_R(p_L, p_R)$ explicitly.
2. Prove that the claim that m is a strictly dominant platform for both candidates is incorrect.
3. Characterize the pure-strategy Nash equilibrium set as a function of (m, τ) .
4. State whether your conclusion changes if the platform space is $\mathbb{R}$.

Exercise 5 (The ultimatum game). Two players divide a unit surplus. Player 1 offers Player 2 a share $x \in X_N = \{0, \frac{1}{N}, \dots, \frac{N-1}{N}, 1\}$. Player 2 accepts or rejects. Acceptance gives payoffs $(1 - x, x)$; rejection gives (d_1, d_2) , where $d_1 + d_2 < 1$. Characterize all subgame-perfect equilibria for arbitrary $N > 2$, paying attention to the responder’s behavior when indifferent. Study the limit as $N \rightarrow \infty$ and explain why replacing the grid by $[0, 1]$ can affect existence or uniqueness.

Exercise 6 (The first-mover (dis-)advantage). For each of the sequential matching-pennies, Cournot–Tullock, and Hotelling–Downs models from class, construct the follower’s complete best-response correspondence before solving for the leader’s action. Determine whether the first mover is advantaged, disadvantaged, or neither relative to simultaneous play. Identify the feature of the follower’s response that produces the comparison and a parameter value at which the conclusion changes, or explain why it never changes.

Exercise 7 (Optional: expected utility under mixing). *In the India–Pakistan Prisoner’s Dilemma from class, use the following mixed strategy profile:*

$$\sigma_{IND}(Nuke) = 0.3, \quad \sigma_{IND}(\neg Nuke) = 0.7, \quad \sigma_{PAK}(Nuke) = 0.7, \quad \sigma_{PAK}(\neg Nuke) = 0.3$$

A proposed solution considers only diagonal outcomes. Explain the error, write the full distribution over action profiles, verify that it sums to one, and compute both expected payoffs. Find the derivative of each expected payoff with respect to India’s probability of choosing Nuke and interpret its sign as a best-response statement.